SCHUBKI NATIONAL AI HACKATHON 2026 · CHALLENGE 01 · ART FIBER SYSTEMS

From customer enquiry to a cited first assessment

An AI assistant for in-line spectroscopy solution design

Wired-Badgers · Mohammed Bashir, Ukasha Khalid, Omema Rizvi

afs-spectroscopy-assistant.vercel.app

THE PROBLEM

Customers know what to monitor, not how to measure it

What the customer brings

- A target: moisture, API content, fat, a contaminant
- A process and a few sentences of context
- Often a premise: “we already have this cable”

What the AFS expert must work out

- Which method: NIR, MIR, Raman, fluorescence, UV/Vis
- Which spectral range, fiber and probe geometry
- Which questions to ask, and which first experiment

> Our assistant drafts that first technical assessment, with every claim linked to its AFS page, for the expert to review, not to replace.

APPROACH

One expert-grade call, with every AFS page in view

All materials in context

Every AFS page, tagged with its source and page number, sits in one cached prompt. A premise check can draw on pages the enquiry never names.

An expert's playbook as the prompt

The role of an AFS application spectroscopist, strict rules, five domain checks and a fixed answer format.

Every claim is clickable

A citation such as S01 p.18 opens the original AFS PDF at that page, so the reviewer checks it in one click.

> *"As simple as possible, as complicated as necessary."*
>
> The principle the AFS experts gave us in the Q&A, applied to the system as well as to its recommendations

ARCHITECTURE

How a request flows through the system

1 · Chat page

Enquiry

The customer’s text, or a follow-up answer

2 · /api/chat

API route

Appends a short reminder of the hard rules

3 · GPT-6.1 via OpenRouter

Agent

One call, medium reasoning effort

4 · Streamed markdown

Assessment

Each citation opens the AFS PDF at its page

System prompt: rules plus all AFS pages, about 69k tokens, cached between requests

Web search, optional: only for facts the materials lack, quoted in a separate section; not used in the final answers

Next.js 16 · Vercel AI SDK · OpenRouter. The command-line runner uses the same agent and logs model, tokens, cost and time per answer.

USE OF THE AFS MATERIALS

From PDFs to page-tagged, citable text

1 · Text layer, copied verbatim

Each page's PDF text, so numbers and units stay exact

2 · Vision notes for what the text misses

Figures, charts and slides read by a vision model: visible values only, unsure readings marked [?]

3 · One tag per page

```
<page id="S01" p="18" href="…pdf#page=18">
```

Special handling

S01 Table 1 (p.6) is scrambled in the PDF text, so the organisers' spreadsheet replaces that page.

Left out on purpose

- The reference document: its worked answers are not evidence; only its answer structure is in the prompt
- The evaluation scenarios: test input only

THE PROMPT

Five domain checks run on every enquiry

Fiber

Does the fiber family carry the needed wavelengths? Checked against S02 p.6.

Geometry

Does the probe suit the sample: clear, opaque, powder, paste, melt? S01 Table 1.

Signal

Is there a physical route from the method to the target as the client defined it?

Calibration

A model is specific to its matrix; a new product needs new reference data.

Reference

Every model needs an independent reference method on the same samples.

Around the checks

Role: an AFS application spectroscopist · five answer types, from full assessment to “outside the supported scope” · a method overview · notes on each source’s scope

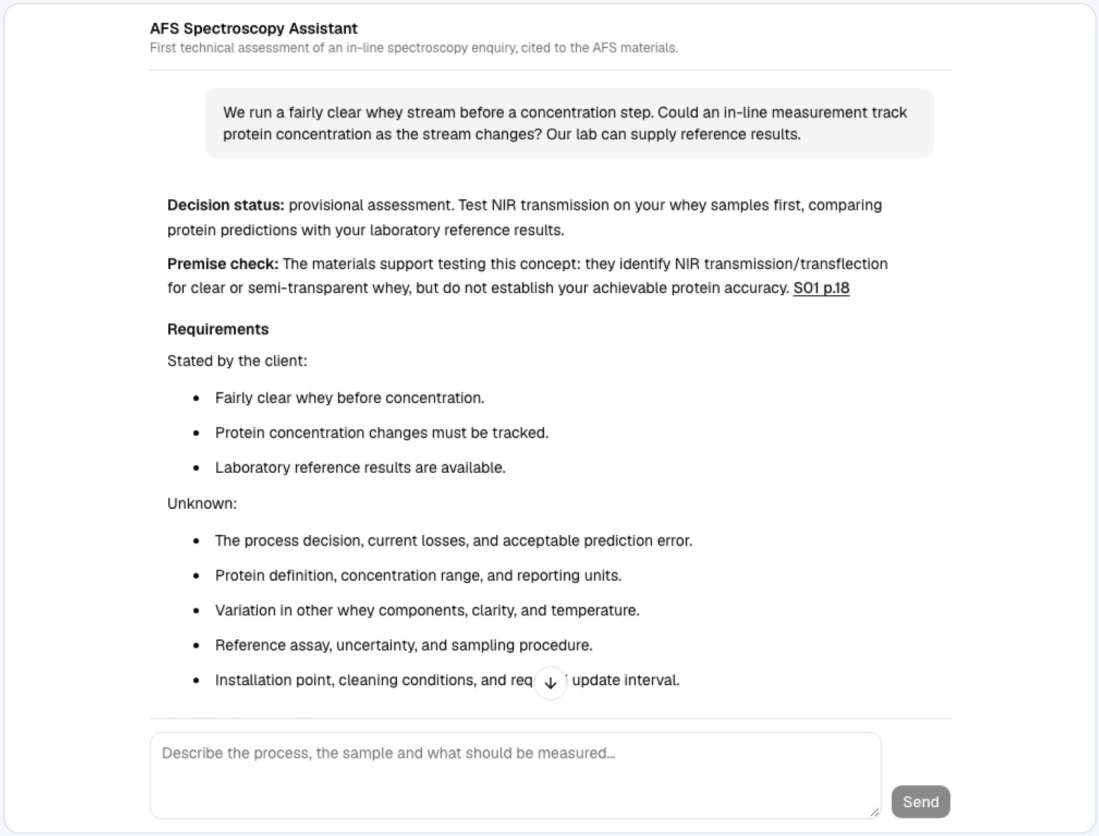
Hard rules

Facts, assumptions and sources kept apart · every number from a cited page or the client · check the premise · respect evidence scope · simplest method first

THE OUTPUT

Every answer leads with the decision

- **Decision status** and what to test first
- **Premise check** with its citation
- Requirements: stated by the client vs. unknown
- Up to 5 questions, each with “Why it matters: if…”
- Recommendation: Method · Why test it · Check first
- Measurement concept, spectral range, risks
- 4-stage feasibility plan and decision basis
- Sources, each a link to its AFS page



The live chat app, answering a whey-stream enquiry

FINAL EVALUATION CASES

In all six cases, the premise is checked before anything is recommended

Case	What the customer assumed or asked	The assistant's verdict
H01	Reuse the NIR calibration; Raman keeps it valid	Premise not supported: test the old model on assayed new-blend samples first
H02	One concept for clear whey and opaque concentrate	Provisional: transmission/transflection for whey, reflection for concentrate
H03	Fluorescence for glucose and viable biomass	Insufficient information: sensitivity alone cannot establish selectivity
H04	Reuse a Vis/NIR silica cable for MIR ATR	Not approved as specified: the silica cable cannot carry mid-IR
H05	Exact sensor for an unnamed contaminant	Insufficient information: compare good and off-spec samples first
H06	In-vitro kidney study as proof for live surgery	No: supports feasibility research, not surgical readiness

Each answer was generated once by the assistant and committed unedited: 9–18 AFS citations each, 5 questions each with a reason, no web sources.

EXAMPLE RECOMMENDATION · H04

Water in petroleum over a reused silica cable

Decision and premise check

Do not approve the concept as specified. MIR ATR merits testing, but the silica cable cannot carry the required MIR wavelengths. [S02 p.6]

Spectral range, worked from the sources

Water region in the petroleum example: 4000–3000 cm^{-1} [S02 p.13], i.e. 2.5–3.33 µm. Vis-NIR silica ends at 2400 nm; chalcogenide spans 1100–6500 nm. [S02 p.6]

One of five questions

What are the link length, route and process conditions?

Why it matters: if the remote link loses too much signal, the analyzer location must change.

Feasibility plan

Lab screening on client samples → ATR model checked against an independent water assay → in-line trial with the real routed link → scale-up and recalibration plan

EVALUATION

We built two systems and let a blind judge choose

9 trap enquiries of our own

Wrong geometry, unsuitable fiber, no signal route, calibration transfer, stretched evidence, out of scope, too vague, plus one sound control, each with a written answer key

An offline check on every output

Status line, premise check, questions with reasons, and every citation pointing to a page that exists

A heavier pipeline (rule engine, verifier, critic, revision) against this one-call design, judged blind in both orders on H01–H06:

Criterion (score of 10)	This system	Heavy pipeline
Technical quality	7.7	6.8
Use of AFS knowledge	7.7	6.0
Feasibility plan	8.0	6.9
Clarification and uncertainty	8.8	6.9
Explainability	8.2	6.2

Preferred in 5 of 6 cases; the heavy pipeline won H05. One LLM judge, so a signal, not proof.

OPERATIONS

Fast and cheap enough for every enquiry

≈ $0.02

per assessment ($0.019–0.021 on H01–H06)

30–35 s

per assessment, streamed as it is written

99.8%

of input tokens served from the prompt cache

All 14 committed answers (H01–H06, D01–D08) cost **$0.29** in total. At this rate, 1,000 assessments a month cost about **$20** in model usage.

Measured per run in runs.json: model, tokens, cached tokens, cost and time.

SCALING

A growing library changes how pages arrive, not how they are cited

Size of the library	How pages reach the model
Up to ~200k tokens (today: 5 documents, ~69k)	Everything in the cached prompt, as now
Hundreds to thousands of pages	A map per document in the prompt; the agent reads pages with a read_pages tool
Millions of pages	Hybrid search picks candidate pages, then the same read_pages tool reads them

A new AFS document: run the ingest once; the page tags and citation links stay the same.

A new expert rule: a line of text in the prompt, not a code change.

OUTLOOK

Honest limits and the next steps

Limits today

- Answers run about 590–680 words, above the ~550 target
- Outputs are checked for structure and valid pages, not claim by claim
- For an undefined contaminant, a broader multi-method screen would be stronger
- Verdicts reviewed by us, not yet by AFS experts; no measured spectra

Next steps with AFS

- Expert corrections become prompt rules and a case library
- Add an offline gate that traces every number and quote to its page
- Send the questionnaire to customers before the first call
- Page maps and a read_pages tool as the library grows

THANK YOU

A first assessment in about 30 seconds, every claim one click from its source

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